



# OLIST CASE

● CUSTOMER RETENTION & RFM SEGMENTATION - OLIST CASE STUDY

PYTHON + POWER BI | CRM / RETENTION / CUSTOMER INSIGHTS

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# BUSINESS PROBLEM

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01

**How can we improve customer retention and repeat purchase?**

02

**Which customer groups should CRM prioritize?**

03

**Where is the biggest reactivation opportunity?**

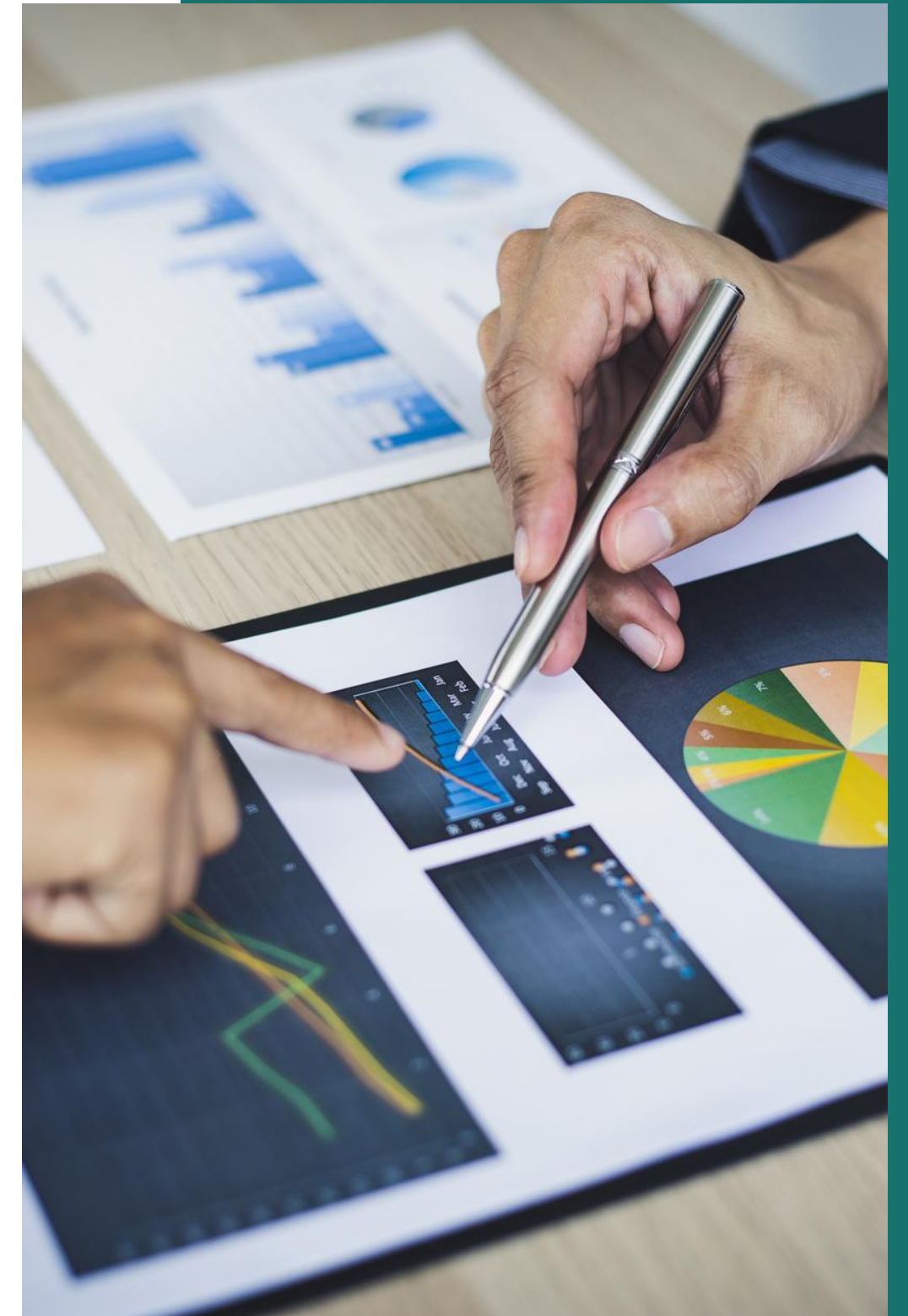
# DATASET AND ANALYTICAL APPROACH

## DATA SET

- Olist Brazilian E-Commerce Public Dataset
- Orders, order items, customers, products

## ANALYTICAL APPROACH

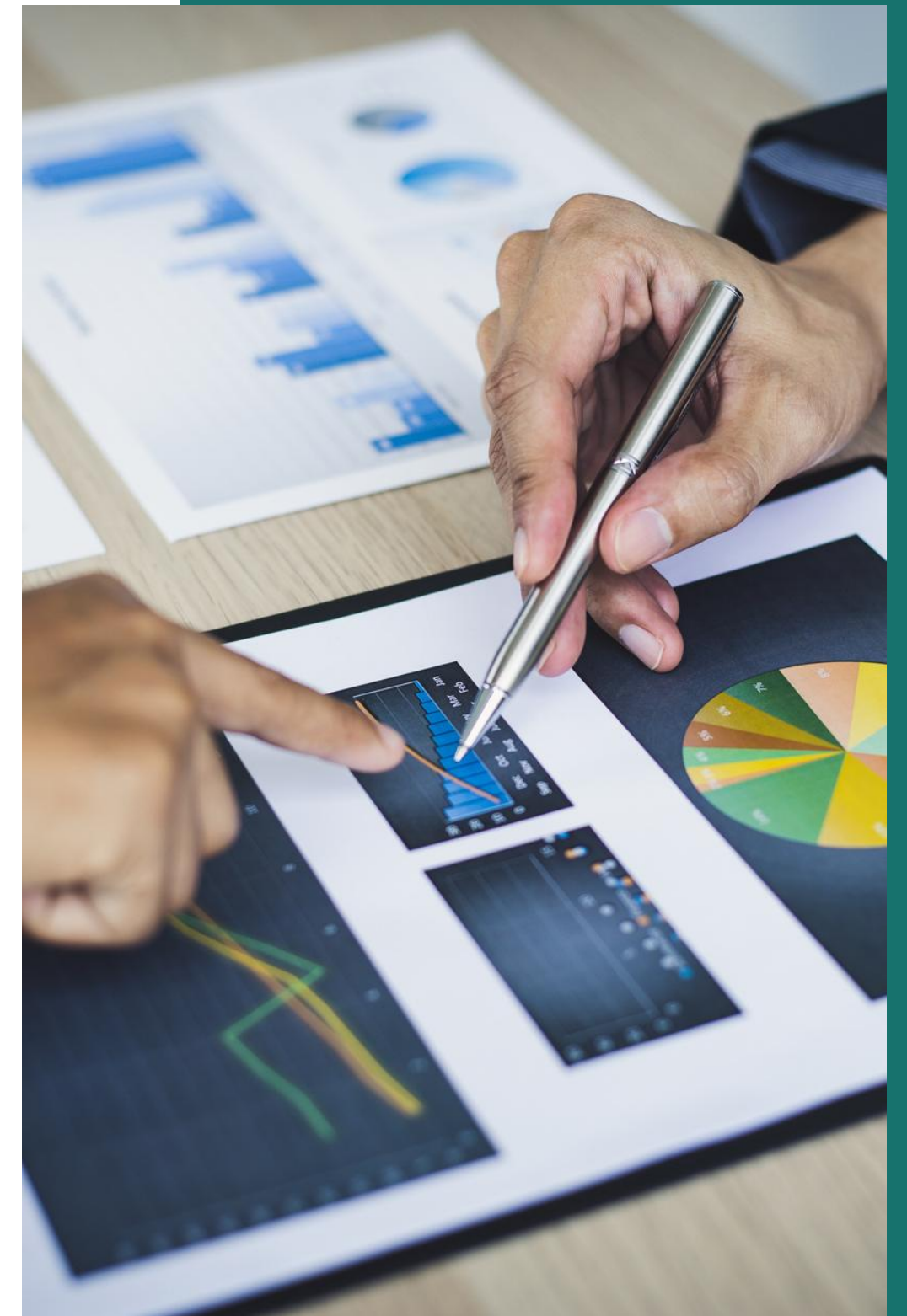
- Data preparation
- Cohort retention analysis
- RFM segmentation
- Power BI Dashboard



# DATA PREPARATION

- Built an order-level fact table
- Calculated revenue per order
- Mapped customer IDs to unique customers
- Exported analytical tables for BI use

RAW TABLES → ORDERS\_FACT → EXPORTS → DASHBOARD





# COHORT RETENTION ANALYSIS

01

Customers grouped by first purchase month

02

Retention measured month by month

03

Output: Cohort retention heatmap

| Customer Retention by Cohort month |         |         |       |       |       |       |       |       |       |       |       |       |       | ⌵ ⌵ ... |
|------------------------------------|---------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|---------|
| cohort_month                       | 0       | 1       | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    | 11    | 12    |         |
| 01/01/2017                         | 100,00% | 0,28%   | 0,28% | 0,14% | 0,42% | 0,14% | 0,42% | 0,14% | 0,14% |       | 0,42% | 0,14% | 0,70% |         |
| 01/01/2018                         | 100,00% | 0,34%   | 0,37% | 0,29% | 0,29% | 0,16% | 0,18% | 0,23% |       |       |       |       |       |         |
| 01/02/2017                         | 100,00% | 0,18%   | 0,31% | 0,12% | 0,43% | 0,12% | 0,25% | 0,18% | 0,12% | 0,18% | 0,12% | 0,31% | 0,12% |         |
| 01/02/2018                         | 100,00% | 0,35%   | 0,40% | 0,30% | 0,25% | 0,22% | 0,21% |       |       |       |       |       |       |         |
| 01/03/2017                         | 100,00% | 0,44%   | 0,36% | 0,40% | 0,36% | 0,16% | 0,16% | 0,32% | 0,32% | 0,08% | 0,36% | 0,12% | 0,20% |         |
| 01/03/2018                         | 100,00% | 0,40%   | 0,30% | 0,30% | 0,12% | 0,12% |       |       |       |       |       |       |       |         |
| 01/04/2017                         | 100,00% | 0,62%   | 0,22% | 0,18% | 0,27% | 0,27% | 0,35% | 0,31% | 0,31% | 0,18% | 0,27% | 0,09% | 0,04% |         |
| 01/04/2018                         | 100,00% | 0,59%   | 0,30% | 0,24% | 0,14% |       |       |       |       |       |       |       |       |         |
| 01/05/2017                         | 100,00% | 0,46%   | 0,46% | 0,29% | 0,29% | 0,32% | 0,41% | 0,14% | 0,26% | 0,26% | 0,26% | 0,35% | 0,23% |         |
| 01/05/2018                         | 100,00% | 0,52%   | 0,26% | 0,18% |       |       |       |       |       |       |       |       |       |         |
| 01/06/2017                         | 100,00% | 0,49%   | 0,40% | 0,43% | 0,30% | 0,40% | 0,36% | 0,23% | 0,13% | 0,20% | 0,30% | 0,36% | 0,16% |         |
| 01/06/2018                         | 100,00% | 0,43%   | 0,27% |       |       |       |       |       |       |       |       |       |       |         |
| 01/07/2017                         | 100,00% | 0,53%   | 0,35% | 0,24% | 0,29% | 0,21% | 0,32% | 0,11% | 0,19% | 0,27% | 0,21% | 0,29% | 0,13% |         |
| 01/07/2018                         | 100,00% | 0,52%   |       |       |       |       |       |       |       |       |       |       |       |         |
| 01/08/2017                         | 100,00% | 0,69%   | 0,35% | 0,27% | 0,35% | 0,52% | 0,30% | 0,27% | 0,15% | 0,15% | 0,25% | 0,20% | 0,12% |         |
| 01/08/2018                         | 100,00% |         |       |       |       |       |       |       |       |       |       |       |       |         |
| 01/09/2016                         | 100,00% |         |       |       |       |       |       |       |       |       |       |       |       |         |
| 01/09/2017                         | 100,00% | 0,70%   | 0,55% | 0,27% | 0,45% | 0,22% | 0,22% | 0,25% | 0,27% | 0,17% | 0,25% | 0,07% |       |         |
| 01/10/2016                         | 100,00% |         |       |       |       |       | 0,38% |       |       | 0,38% |       | 0,38% |       |         |
| 01/10/2017                         | 100,00% | 0,72%   | 0,25% | 0,09% | 0,23% | 0,21% | 0,21% | 0,37% | 0,28% | 0,18% | 0,21% |       |       |         |
| 01/11/2017                         | 100,00% | 0,57%   | 0,37% | 0,17% | 0,17% | 0,18% | 0,11% | 0,18% | 0,13% | 0,06% |       |       |       |         |
| 01/12/2016                         | 100,00% | 100,00% |       |       |       |       |       |       |       |       |       |       |       |         |
| 01/12/2017                         | 100,00% | 0,21%   | 0,28% | 0,34% | 0,26% | 0,21% | 0,17% | 0,02% | 0,19% |       |       |       |       |         |

# RFM SEGMENTATION

**Recency** = days since last purchase  
Lower is better

**Monetary** = total customer revenue  
Higher is better

**Frequency** = number of orders  
Higher is better

## Segments:

CHAMPIONS

LOYAL CUSTOMERS

POTENTIAL LOYALISTS

NEW CUSTOMERS

AT RISK

HIBERNATING

NEEDS ATTENTION

# DASHBOARD OVERVIEW

Customer Retention by Cohort month

| cohort_month | 0       | 1       | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    | 11    | 12    |
|--------------|---------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 01/01/2017   | 100,00% | 0,28%   | 0,28% | 0,14% | 0,42% | 0,14% | 0,42% | 0,14% | 0,14% |       | 0,42% | 0,14% | 0,70% |
| 01/01/2018   | 100,00% | 0,34%   | 0,37% | 0,29% | 0,29% | 0,16% | 0,18% | 0,23% |       |       |       |       |       |
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# DASHBOARD OVERVIEW

93 Mil

Total Customers

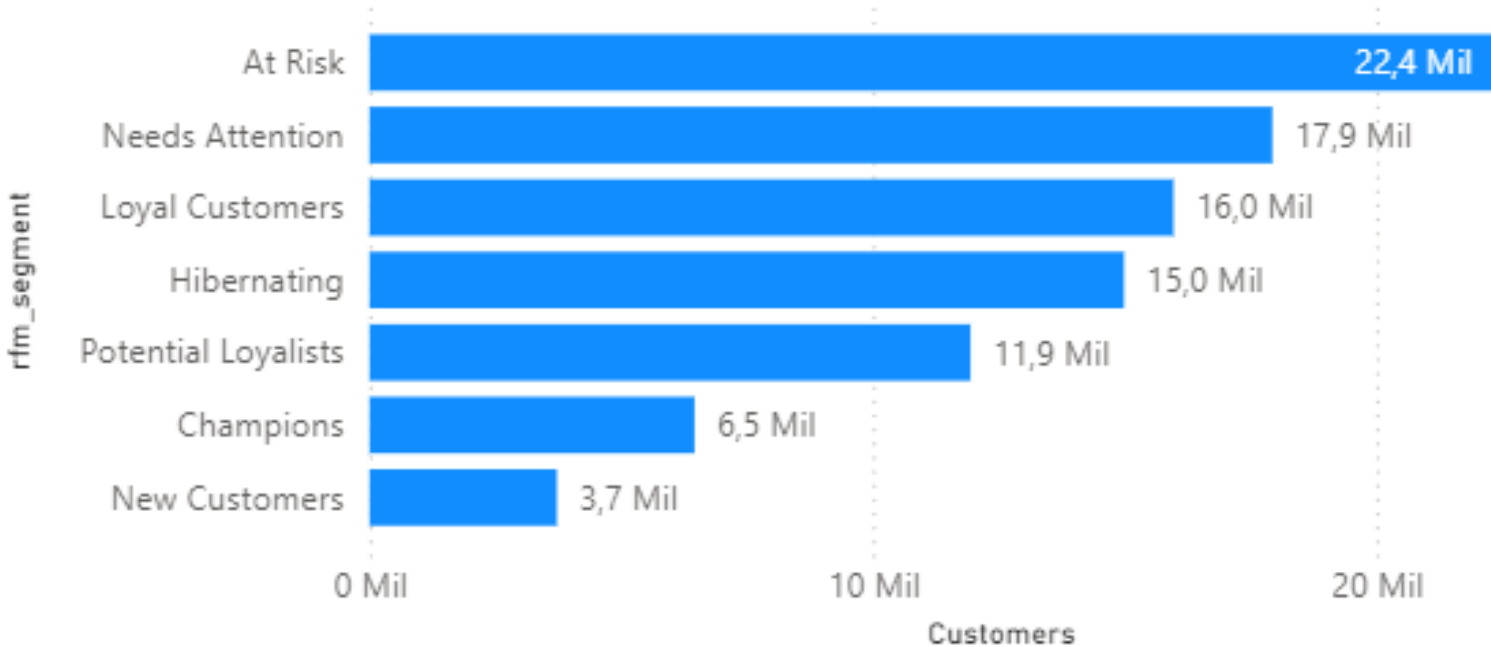
15,42 Mi

Total Revenue

At Risk

Top Revenue Segment

Customers by segment



Revenue by segment



Segment Summary

| rfm_segment         | customers | avg_recency_days | avg_frequency | avg_monetary | total_revenue  |
|---------------------|-----------|------------------|---------------|--------------|----------------|
| At Risk             | 22357     | 393,94           | 1,05          | 166,90       | \$3.731.464,03 |
| Needs Attention     | 17934     | 180,18           | 1,00          | 135,89       | \$2.437.105,48 |
| Loyal Customers     | 15969     | 150,72           | 1,05          | 115,58       | \$1.845.762,01 |
| Hibernating         | 14986     | 395,51           | 1,00          | 162,94       | \$2.441.759,58 |
| Potential Loyalists | 11937     | 90,95            | 1,00          | 196,94       | \$2.350.825,53 |
| Champions           | 6455      | 90,55            | 1,18          | 312,13       | \$2.014.823,09 |
| New Customers       | 3720      | 45,95            | 1,00          | 160,76       | \$598.034,03   |

Recommended CRM Actions by Segment

| rfm_segment         | goal                                      | primary_action                  | timing     |
|---------------------|-------------------------------------------|---------------------------------|------------|
| At Risk             | Win back valuable lapsed customers        | Reactivation campaign           | 30–60 days |
| Champions           | Retain and grow top customers             | Loyalty, referral, exclusives   | Ongoing    |
| Hibernating         | Selectively reactivate inactive customers | Strong offer and reminder       | 60+ days   |
| Loyal Customers     | Increase frequency and basket size        | Cross-sell, bundles, engagement | 15–30 days |
| Needs attention     | Prevent further drop-off                  | Light engagement/reminder       | 15–30 days |
| New customers       | Drive second purchase                     | Welcome/onboarding journey      | 7–14 days  |
| Potential Loyalists | Convert into loyal repeat buyers          | Second-order incentive          | 7–21 days  |



# KEY INSIGHTS

**AT RISK** is the largest and highest-revenue segment

**HIBERNATING** still represents meaningful reactivation potential

**CHAMPIONS** show the highest average customer value

**COHORT RETENTION** drops quickly after the first months

Segment Summary

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# RECOMMENDATIONS, LIMITATIONS, AND CLOSING

Recommended CRM Actions by Segment

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|---------------------|-------------------------------------------|---------------------------------|------------|
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## Next Steps:

SECOND PURCHASE RATE WITHIN  
90 DAYS

CUMULATIVE REPEAT  
PURCHASE BY COHORT

CATEGORY-LEVEL  
ANALYSIS



# THANK YOU

● FOR YOUR ATTENTION